GPT-2 XL: Warsaw → Poland → Polish

Poland · other countries · Polish · other languages

“The most widely spoken language of the country whose capital is Warsaw is”

J-lens

layer	1	2	3	4	5	6	7	8	9	10
18	...	also	called	....	.....	.	..	named	is	·
20	also	called	nevertheless	named	therefore	.	....	...	.....	now
22	called	also	named	....	now	nevertheless	...	.	.....	nowadays
24	now	nowadays	called	named	also	nicknamed	nevertheless	therefore	itself	name
26	Sanskrit	English	also	called	now	nowadays	named	Welsh	languages	Hebrew
28	Sanskrit	English	Welsh	lingu	languages	Gael	Hungarian	Bulgarian	Spanish	Languages
30	**Polish**	languages	Languages	Sanskrit	Welsh	Hungarian	lingu	anguage	Bulgarian	Gael
32	**Polish**	Lithuan	Hungarian	**Poland**	Romanian	Bulgarian	Slav	Serbian	Sloven	Czech
34	**Polish**	**Poland**	Hungarian	Lithuan	**Poles**	Slav	Romanian	Belarus	Czech	Bulgarian
36	**Polish**	**Poland**	Hungarian	Lithuan	**Poles**	Slav	Czech	Warsaw	languages	Romanian
38	**Polish**	**Poland**	Lithuan	**Poles**	Hungarian	Czech	Warsaw	Romanian	Slav	Belarus
40	**Polish**	**Poland**	Lithuan	**Poles**	Hungarian	Warsaw	Czech	Romanian	Slov	Belarus
42	**Polish**	**Poland**	Lithuan	Hungarian	**Poles**	Czech	Slov	Belarus	Romanian	Warsaw
44	**Polish**	Czech	Lithuan	Hungarian	**Poland**	**Poles**	Slov	Belarus	Esper	Ukrainian
46	**Polish**	Czech	Lithuan	English	**Poland**	Hungarian	Hebrew	Ukrainian	German	Belarus
47 (output)	**Polish**	now	also	the	English	not	in	German	a	Russian

rank at the last prompt token

logit lens

layer	1	2	3	4	5	6	7	8	9	10
18	undoubtedly	currently	reportedly	located	likely	situated	now	attracting	hardly	probably
20	undoubtedly	ranked	located	reportedly	attracting	also	situated	dominated	now	hardly
22	ranked	now	called	reportedly	undoubtedly	dubbed	named	nicknamed	also	currently
24	undoubtedly	now	named	dubbed	reportedly	nicknamed	called	ranked	currently	listed
26	English	Gael	Sanskrit	Azerbai	Arabic	now	undoubtedly	Welsh	Finnish	Hebrew
28	English	Sanskrit	Arabic	Gael	Hebrew	Spanish	undoubtedly	Bulgarian	German	Welsh
30	English	**Polish**	Spanish	Serbian	Hebrew	Bulgarian	German	languages	Sanskrit	Arabic
32	**Polish**	English	Bulgarian	Serbian	Lithuan	German	Ukrainian	Hungarian	Romanian	**Poland**
34	**Polish**	English	German	Hungarian	**Poland**	Bulgarian	Ukrainian	Serbian	Russian	Lithuan
36	**Polish**	English	**Poland**	Hungarian	Lithuan	Czech	Slav	Warsaw	German	Ukrainian
38	**Polish**	**Poland**	English	Czech	Hungarian	Warsaw	Lithuan	German	Ukrainian	Slav
40	**Polish**	**Poland**	English	German	Czech	Hungarian	Lithuan	Warsaw	Ukrainian	Russian
42	**Polish**	English	German	**Poland**	Czech	Hungarian	Russian	Ukrainian	Lithuan	Hebrew
44	**Polish**	English	German	Czech	Russian	**Poland**	Hungarian	Ukrainian	now	Lithuan
46	**Polish**	English	now	also	the	not	German	Russian	in	Czech
47 (output)	**Polish**	now	also	the	English	not	in	German	a	Russian

rank at the last prompt token

Gemma 4 31B: largest country → Russia → Moscow

Russia · other countries · Moscow (any language) · other capitals

“Fact: The capital city of the largest country by area is”

J-lens

layer	1	2	3	4	5	6	7	8	9	10
22	No	Bus	No	BUS	Bush	BUS	Bus	NO	bus	Buz
24	No	BUS	Bush	No	Bus	<eos>	.[	<!	]]	...(
26	<eos>	,...	</h4>	↵	,...	...	Serbia	...(	</h5>	
28	.[	Serbia	**Russian**	**Russia**	Kyrgyzstan	famous	only	called	Belarus	Leningrad
30	**Russia**	Mongolia	Serbia	Leningrad	**Russian**	Afghanistan	Uzbekistan	Kyrgyzstan	**Moscow**	Ukraine
32	Mongolia	**Russia**	Afghanistan	Kyrgyzstan	Serbia	Uzbekistan	Turkmenistan	Kyrgyz	countries	Azerbaijan
34	Mongolia	**Russia**	Kyrgyzstan	Turkmenistan	Uzbekistan	Afghanistan	Kyrgyz	Serbia	Belarus	Kazakhstan
36	Mongolia	**Russia**	Kyrgyzstan	Turkmenistan	Uzbekistan	Kazakhstan	Belarus	**Moscow**	Azerbaijan	Lithuania
38	Mongolia	**Russia**	Uzbekistan	Turkmenistan	Antarctica	**Moscow**	Kazakhstan	**Russia**	Kyrgyzstan	neither
40	also	ALSO	тоже	**Moscow**	neither	también	Beijing	также	located	також
42	**Moscow**	**Москва**	Beijing	Brasília	東京都	**Moscow**	Tehran	Istanbul	Jakarta	Beijing
44	**Москва**	Brasília	**Moscow**	Beijing	北京	Islamabad	東京都	首都	北京市	**Moscow**
46	**Moscow**	**Москва**	Beijing	Brasília	**Moscow**	北京	Islamabad	Jakarta	**Москвы**	北京市
48	**Moscow**	**Москва**	Beijing	Islamabad	Brasília	**Moscow**	Jakarta	Canberra	**Москвы**	**Moscou**
50	**Moscow**	**Moscow**	**Москва**	Beijing	Brasília	**Moscou**	Islamabad	**Москвы**	Jakarta	**Москве**
52	**Moscow**	**Moscow**	**Москва**	**Moscou**	**Москвы**	**Москве**	**Moskau**	**Москва**	**Моск**	Islamabad
54	**Moscow**	**Moscow**	**Moscou**	**Москва**	**Moskau**	**Москвы**	**Москве**	**Москва**	Beijing	**Моск**
56	**Moscow**	**Moscow**	**Moscou**	**Москва**	**Moskau**	**Москвы**	**Mosc**	**Москве**	**Моск**	**Russia**
58	**Moscow**	Canberra	Beijing	**Moscow**	Islamabad	Brasília	**Moscou**	Astana	**Russia**	Jakarta
59 (output)	**Moscow**	also	the	not	located	a	␣	Beijing	**Russia**	in

rank at the last prompt token

logit lens

layer	1	2	3	4	5	6	7	8	9	10
22	最早	เรา	க்ஸாண்ட	砲	amalgamation	Ри	𓅓𓅓	ABSORBED	missions	ла
24	При	Ри	mittance	providing	ла	У	premios	достаточно	лете	মই্দ
26	Очень	মই্দ	Ри	расходы	лете	дуже	Eigenschafte	phenylsulfan	РИ	非常
28	(	ArgsEnv	?!?!	(!)	ReferencePos	Unlike	(!)	MakeRange	Unlike	UsedError
30	that	which	(	is	𐌺	While	উত্তরঃ	would	during	অঞ্চ
32	that	which	(	would	d	t	n	c	b	w
34	that	which	would	d	c	w	(	the	g	n
36	that	which	c	d	While	the	would	t	w	While
38	not	that	While	While	which	either	c	also	d	the
40	also	також	not	also	Also	também	тоже	также	również	också
42	also	not	also	While	When	Not	While	more	тоже	Also
44	also	not	also	столи	北京	首都	тоже	Not	While	more
46	also	北京	not	首都	столи	Brasília	東京都	also	北京市	тоже
48	not	北京	c	also	t	also	s	more	Not	столи
50	北京	**Moscow**	not	also	also	not	Not	**Москва**	t	**Москва**
52	**Moscow**	**Moscow**	**Москвы**	**Москва**	**Moscou**	**Москве**	**Моск**	**Москва**	北京	**Mosk**
54	**Moscow**	**Moscow**	**Moscou**	**Москва**	**Москвы**	**Москве**	**Моск**	**Moskau**	**Москва**	**Mosc**
56	**Moscow**	**Москва**	**Moscow**	**Moscou**	**Москвы**	**Москве**	**Моск**	**Moskau**	**Mosc**	**Mosca**
58	**Moscow**	**Moscow**	**Moscou**	Canberra	**Москва**	**Москве**	**Moskau**	موس	**Моск**	Beijing
59 (output)	**Moscow**	also	the	not	located	a	␣	Beijing	**Russia**	in

rank at the last prompt token

Gemma 4 31B: country shaped like a boot → Italy → Rome

Italy · other countries · Rome (any language) · other cities

“The capital of the country shaped like a boot is”

J-lens

layer	1	2	3	4	5	6	7	8	9	10
22	No	BUS	Bus	No	Bush	NO	bus	Buz	BUS	.
24	No	Bush	BUS	Bus	No	BUS	NO	bus	Bus	Ho
26	yummy	called	whopping	chilly	amazingly	crazy	fabulous	chubby	funny	bigger
28		**Italy**	called	famous	yummy	whopping	amazingly	hardly	gorgeous	<b>
30		<b>	**Italy**		yummy	famous	pretty	whopping	amazingly	called
32	**Italy**	famously	Spain	England	yummy	famous		....	Romania	
34	**Italy**		Istanbul	Spain	Constantinop	France	famously	Romania	**Italy**	cities
36		**Italy**	Istanbul	Constantinop	famously	yummy	Spain	cities	famous	
38	_____	_____		_____	____	___	_____	_____	_____	_____
40	_____	.....	...?	.....	· · ·	.....	...?	.....	.....	_____
42	_____	· · ·	.....	.....	...?	.....	.....	.....	.....	_____
44	_____	_____	_____	___	_____	_____	_____	__:	_____	· · ·
46	_____	_____	Москва	_____	___	**Italy**	**Rome**	_____	_____	首都
48	**Rome**	_____	**Italy**	_____	_____	___	_____	_____	_____	**Италии**
50	**Rome**	**Rome**	_____	**Italy**	_____	___	_____	Bucharest	· · ·	Bucure
52	**Rome**	**Rome**	**Roma**	罗马	Roman	ローマ	**Рим**	**Italy**	**Roma**	**ROME**
54	**Rome**	**Rome**	**Roma**	罗马	**Italy**	**Рим**	**ROME**	**rome**	**Roma**	ローマ
56	**Rome**	_____	___	**Rome**	_____	_____	_____	**Roma**	____	_____
58	**Rome**	**Rome**	**Italy**	罗马	**Roma**	located	**rome**	brimming	**ROME**	_____
59 (output)	**Rome**	a	the	␣	one	located	:	also	known	not

rank at the last prompt token

logit lens

layer	1	2	3	4	5	6	7	8	9	10
22	ла	на	расходы	premios	最早	érieures	πίνα	glittering	เรา	áme
24	最早	ла	расходы	premios	Anosov	newMovie	shipping	érieures	При	identité
26	Caedwalla	glittering	расходы	protecting	增加了	три	·	可以在	ylmethylsulf	increases
28	While	which	Unlike	в	增加了	ArgsEnv	Unlike	because	?!?	훨씬
30	While	that	would	which	when	When	𐌺	would	ट्र	That
32	that	which	when	While	would	For	That	When	तौर	the
34	which	that	the	would	While	For	when	with	d	c
36	that	the	When	when	what	c	which	While	d	When
38	When	what	that	not	While	What	While	what	What	the
40	also	not	aptly	While	undoubtedly	exceedingly	When	intricately	również	While
42	not	also	one	When	While	a	While	an	Not	renowned
44	also	not	one	renowned	столи	a	an	famed	Not	undoubtedly
46	not	столи	also	renowned	one	famed	Not	a	exceedingly	bustling
48	not	**Rome**	столи	c	famed	bustling	**Rome**	t	exceedingly	renowned
50	**Rome**	**Rome**	not	北京	____	without	_____	bustling	notoriously	not
52	**Rome**	**Rome**	**Roma**	罗马	**ROME**	ローマ	Romans	Roman	romanos	röm
54	**Rome**	**Rome**	**Roma**	罗马	**rome**	ローマ	**Рим**	**Italy**	**Roma**	**ROME**
56	**Rome**	**Rome**	罗马	ローマ	**Roma**	**ROME**	**rome**	**рим**	**Рим**	____
58	**Rome**	**Rome**	罗马	**Roma**	**ROME**	ローマ	**Italy**	रोम	**Рим**	**روم**
59 (output)	**Rome**	a	the	␣	one	located	:	also	known	not

rank at the last prompt token

Top-10 vocabulary tokens per layer at the last prompt token; the last row is the model's own output, and every model answers correctly. WikiText J-lenses.